Code No: 134BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, March - 2022
FORMAL LANGUAGES AND AUTOMATA THEORY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Construct a DFA accepting the set of all strings ending with 'bb' over Σ={a,b}.
b) Briefly discuss about Finite Automata with Epsilon- Transition.
c) Draw the transition diagram of a FA which accepts all strings of 0's and 1's in which the number of 0's are odd and 1's are even? [5+5+5]

2.a) Show that the language L = {ww | w ε {a, b}*} is not regular.
b) Write the steps in minimization of FA.
c) Construct finite automata for the regular expression (1*0 +10*). [5+5+5]

3.a) Design Push Down Automata for the language
$L=\{ww^R \mid w \varepsilon (0+1)^*\}$.
b) Convert the following grammar into a PDA that accepts the language by empty stack
$S \rightarrow 0S1|A$
$A \rightarrow 1A0 |S| \varepsilon$ [7+8]

4.a) Eliminate ε -productions from the grammar G given as
$A \rightarrow aBb \mid bBa$
$B \rightarrow aB \mid bB \mid \varepsilon$
b) Design Turing Machine which will recognize strings containing equal number of 0's and 1's. [7+8]

5.a) Convert the following grammar to Greibach Normal Form : $S \rightarrow SS \mid 0S1 \mid 01$.
b) Explain Decision properties of Context Free Languages. [8+7]

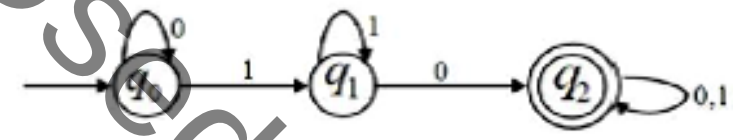
6. Write about the following:
a) Linear-Bounded Automata
b) Recursively enumerable language
c) Decidability of PCP. [5+5+5]

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7.a) Let $\Sigma = \{0,1\}$ and A,B be the list of 3 strings each. Verify if PCP given below has a solution or not?

	List A	List B
1	wi	xi
1	00	0
2	001	11
3	1000	011

b) Construct the Regular expression of the following Finite Automata: [7+8]



8.a) Explain about the identity rules of Regular Expressions.

b) Construct the CFG for the PDA $M = (\{q_0,q_1\}, \{0,1\}, \{R,Z_0\}, \delta, q_0, Z_0, \Phi)$ and δ is given by:

$\delta(q_0,1,Z_0)=(q_0,RZ_0)$
$\delta(q_0,1,R)=(q_0,RR)$
$\delta(q_0,0,R)=(q_1,R)$
$\delta(q_1,0, Z_0)=(q0, Z0)$
$\delta(q_0,\varepsilon, Z_0)=(q_0, \varepsilon)$
$\delta(q_1,1,R)=(q_1,\varepsilon)$. [7+8]

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